

ASSESSMENT-1

TOOL

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Slot :- B

Course code :- MIA0401

Course :- deep learning.

46
50

92%
ms

A coin is tossed 10 times & the outcome are
HHTHHHTHHT

MIE is a statistical method used to estimate
the value of unknown parameters from data.

Formula :-

$$p^{\wedge} = \frac{\text{numbers of successes}}{\text{Total no. of trials}}$$

Total tosses = 10

no. of Heads = 7

Calculation:-

$$\begin{aligned} p^{\wedge} &= \frac{7}{10} \\ &= 0.7 \end{aligned}$$

Answer:-

$$p = 0.7$$

- 2) out of 50 student, 42 passed the exam MLE estimates the probability of success by dividing the no. of successful.

$$\hat{p} = \frac{\text{no. of successes}}{\text{Total observations}}$$

Given

Total Student = 50

Passed student = 42.

Calculation :-

$$\hat{p} = \frac{42}{50}$$

$$\hat{p} = 0.84$$

Answer - 0.84

Conclusion :- student passes the exam is 84.

- 3) MLE estimate the probability of an event using the observed sample data.

$$\hat{p} = \frac{\text{no. of defective bulbs}}{\text{Total Bulbs}}$$

Given,

$$\text{Total Bulbs} = 100$$

$$\text{Bulbs} = 8$$

Calculation:-

$$\hat{p} = \frac{8}{100}$$

$$\hat{p} = 0.08.$$

Answer:-

The estimated probability that a bulb is defective is 0.08 (8%)

MLE estimate the probability of an event by dividing the no. of successful event by the total no. of observation.

$$\hat{p} = \frac{\text{no. of success}}{\text{Total trials}}$$

Given

60 = Total Rolls.

18 = no. of sixes.

calculation :-

$$\frac{18}{60}$$

$$= 0.3$$

probability of getting 6 is 0.3 (30%).

Given,

$$x = 4$$

$$w = 2$$

Actual output = 10

$$\eta = 0.01$$

$$\text{Gradient} = 4$$

(i) predicted output

$$y = w \times x$$

$$y = 2 \times 4$$

$$\boxed{y = 8}$$

(ii) Error

$$\text{Error} = \text{Actual} - \text{predicted output}$$

$$= 10 - 8$$

$$= 2$$

$$(iii) \text{ New weight} = w - (\eta \times \text{gradient})$$

$$= 2 - (0.01 \times 4)$$

$$= 1.96$$

md